Comprehensive Literature Review: Hemodynamic Modeling of Arterial Bifurcations, Stenosis Severity, Blood Rheology, Pulsatility, and Stent-Assisted Restoration

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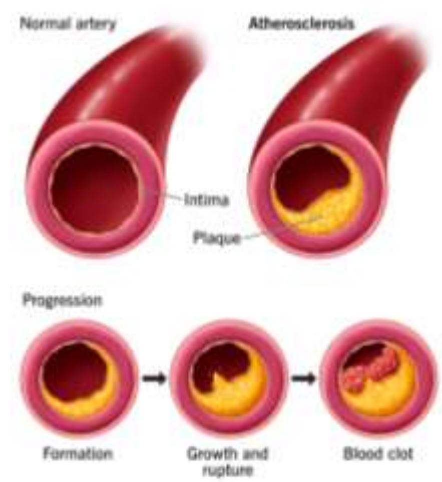
1. Introduction and Pathophysiology of Arterial Disease

Cardiovascular diseases, specifically those involving arterial narrowing, remain a primary focus of biomechanical and medical device research. To understand the operational characteristics of therapeutic interventions, it is critical to distinguish between fundamental arterial pathologies. Arteriosclerosis is a generalized condition characterized by the thickening and loss of elasticity of arterial walls, rendering them stiff and non-compliant. Atherosclerosis, a specific sub-type of arteriosclerosis, involves the localized focal accumulation of lipids, cholesterol, and cellular debris within the intimal layer, forming what is clinically referred to as fibro-fatty plaque. This progression typically proceeds through three distinct phases: plaque formation, growth and eventual rupture, and subsequent acute blood clot or thrombus formation. The clinical consequence of atherosclerosis is arterial lumen narrowing (stenosis), which forces a drastic modification of local hemodynamics, altering blood flow patterns and increasing mechanical stresses that further exacerbate endothelial damage.



2. Bifurcation Lesion Classification Systems

Arterial bifurcations represent geometrically complex regions highly predisposed to plaque deposition due to disrupted flow structures. To standardize descriptions of stenosis distribution within bifurcated segments, several anatomical and procedural classification frameworks are utilized in clinical practice and computational modeling:

2.1 Medina Classification System

The Medina classification is a straightforward binary system used to define the spatial distribution of **significant stenotic lesions (defined as a lumen diameter reduction of 50% or more)** across three specific bifurcated segments: the Proximal Main Vessel (PMV), the Distal Main Vessel (DMV), and the Side Branch (SB). The classification uses a comma-separated format (PMV, DMV, SB) where **'1' denotes significant stenosis** and **'0' indicates its absence.**

Code	Description
1,0,0	Proximal main vessel only
0,1,0	Distal main vessel only
0,0,1	Side branch only
1,1,0	Both main vessel segments
1,0,1	Proximal main vessel and side branch
0,1,1	Distal main vessel and side branch
1,1,1	All three segments

2.2 MADS Classification System

In contrast to purely anatomical classifications, the MADS system is orientational and procedural, helping interventionalists **define initial and subsequent stenting strategies** based on the primary vessel targeted. The acronym corresponds to: Main vessel primary stenting (M), Across side branch ostium stenting (A), Double-planned two-stent deployment strategies (D), and Side branch primary stenting first (S). It directly guides the optimal deployment sequence of vascular restoration devices.

Classification	Initial Strategy	Further Management	Priority
M – Main	Stent in main vessel	Extend if needed	Main vessel primary target
A – Across	Across side branch ostium	Treat SB only if compromised	Main vessel first
D – Double	Planned two-stent strategy	Both branches stented	Equal importance
S – Side	Stent in side branch first	Treat main vessel later if needed	Side branch primary target

2.3 Movahed Classification System

The Movahed system presents a highly comprehensive string format:

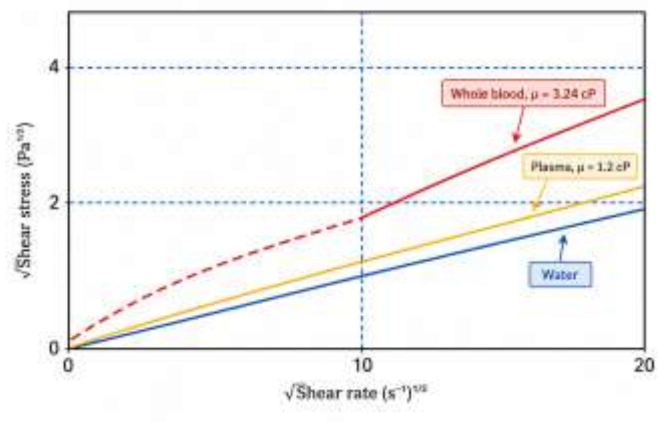
B + (Suffix 1) + (Suffix 2) + (Suffix 3) + (Suffix 4), which integrates anatomy, branch size significance (Uppercase for large, lowercase for small), bifurcation angle relative to a 70-degree threshold, and specific complexity modifiers such as calcification (CA), thrombus (T), or left main (LM) involvement. This method provides the detailed multi-parametric resolution needed for patient-specific computational fluid dynamics (CFD) geometry development.

Suffix	Category	Code	Meaning
Suffix 1	Location/Vessel Size	C	Close to bifurcation
		N	Nonsignificant side branch
		S	Small proximal vessel
		L	Large proximal vessel
Suffix 2	Diseased Ostium	1M	Main branch only
		1S	Side branch only
		2	Both branches involved
Suffix 3	Bifurcation Angle	V	<70°
		T	>70°
Suffix 4	Special Features	CA	Calcified lesion
		LM	Left main involvement

3. Rheological Behavior of Blood

Blood is a heterogeneous suspension of cellular elements (erythrocytes, leukocytes, platelets) suspended within a liquid plasma matrix. While plasma behaves as a purely Newtonian fluid with a stable dynamic viscosity of approximately 1.2 to 1.3 mPa·s, whole blood exhibits non-linear, non-Newtonian behaviors governed by red blood cell (RBC) macro-dynamics:

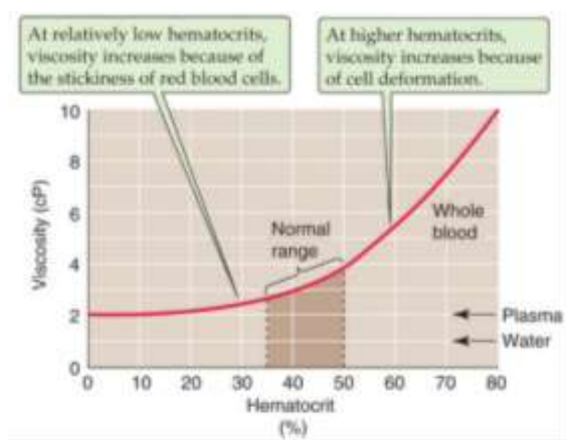
- **Shear-Thinning Behavior: At low shear rates (< 1 s^{-1}):** low hydrodynamic forces allow RBCs to aggregate into linear structures known as rouleaux, significantly increasing apparent viscosity. At intermediate shear rates (1–100 s^{-1}), these aggregates are systematically disrupted. At high shear rates (> 100 s^{-1}), RBCs undergo individual elastic deformation and align parallel to fluid streamlines, causing viscosity to reach an asymptotic low-shear baseline.



- **Yield Stress Phenomenon:** Due to weak cell-to-cell attractive forces and continuous rouleaux networking, blood exhibits a finite threshold stress required to initiate motion. This parameter is highly critical in microcirculation, low-velocity post-stenotic recirculation zones, and stagnation points.

- **Viscoelasticity:** Erythrocyte membranes store elastic energy under stress and exhibit time-dependent shape recovery, resulting in frequency-dependent viscosity and a phase lag between stress and strain during pulsatile flow.

- **Hematocrit Dependence:** Viscosity scales strictly with hematocrit (typically 40–45% physiologically). Elevated cell concentrations dramatically accelerate cell collision frequency, enhance rouleaux networks, and compound total flow resistance.



4. Mathematical Formulations of Rheological Viscosity Models

To capture these shear-dependent viscosity variations in numerical environments like ANSYS Fluent, four constitutive formulations are widely analyzed:

4.1 The Carreau Model

The Carreau model is a multi-parameter, highly non-linear formulation that continuously maps viscosity across the entire physiological shear range without abrupt discontinuities:

$$\mu(\dot{\gamma}) == \mu_\infty + (\mu_0 - \mu_\infty)[1 + (\lambda\dot{\gamma})^2]^{\frac{n-1}{2}}$$

Where:

μ_0: represents the zero-shear viscosity plateau (aggregation dominant)

μ_∞: represents the infinite-shear Newtonian limit (fully aligned plasma state)

λ: is a characteristic relaxation time constant governing the transition zone

n: is the dimensionless flow behavior index representing the degree of shear-thinning ($n < 1$).

This model provides excellent validation against experimental blood data across arterial bifurcations under pulsatile conditions.

4.2 The Cross Model

The Cross formulation also enforces asymptotic limits at zero and infinite shear using an alternative mathematical architecture:

$$\mu(\dot{\gamma}) = \mu_0 + \frac{\mu_0 - \mu_\infty}{1 + (k\dot{\gamma})^m}$$

Where m is a dimensionless exponent controlling the transitional decay slope. While computationally efficient for broad engineering approximations, it lacks the detailed microstructural flexibility of the Carreau model when fitting biological flow properties across highly dynamic arterial transitions.

4.3 The Power Law Model

A simplified two-parameter empirical model mapping viscosity directly via an exponential relation:

$$\mu(\dot{\gamma}) = \mathrm{K}\,\dot{\gamma}^{n-1}$$

Where: **K** is the consistency index.

This formulation lacks zero- and high-shear plateaus, predicting unphysical infinite viscosities as shear rates approach zero and underestimating viscosity in high-shear regimes. It is strictly limited to preliminary parametric assessments over narrow intermediate shear bands.

4.4 The Casson Model

A specialized yield-stress model designed to govern low-shear flow thresholds:

$$\sqrt{\tau} = \sqrt{\tau_y} + \sqrt{\eta\,\dot{\gamma}}$$

Where τ_y represents the activation yield stress and η is the core plastic viscosity. This model accurately predicts solid-like behavior below threshold limits, making it indispensable for modeling thrombosis risks and localized stagnation fields behind severe stenotic throats.

5. Inlet Boundary Waveforms and Spatial Profiles

Hemodynamic simulation fidelity is heavily dictated by temporal and spatial boundary constraints. Common profiles include:

- **Constant (Steady) Waveforms:** Time-invariant models ($\partial v/\partial t = 0$) that serve as initial baseline benchmarks for computational validation and debugging.
- **Pulsatile (Physiological) Waveforms:** Time-dependent profiles mirroring the native cardiac cycle. These divide strictly into a rapid systolic contraction phase (producing maximum acceleration and peak velocity) and a prolonged diastolic relaxation phase, often punctuated by a dicrotic notch resulting from aortic valve closure.

Key Phases:

- **Systole:** The heart contracts, producing a peak in velocity
- **Diastole:** The heart relaxes, causing velocity to decrease
- **Dicrotic Notch (optional):** A small fluctuation due to valve closure

Physical Meaning: This waveform reflects real physiological blood flow, including acceleration and deceleration phases.

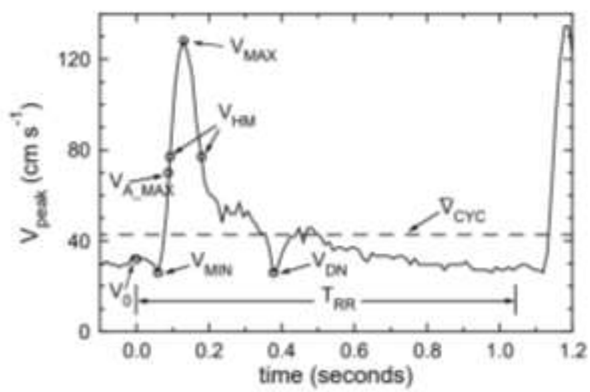


Table 1. Definitions of peak-velocity waveform feature points.

Parameter	Definition
N_{CYC}	Number of cardiac cycles in one acquisition sequence
T_{RR}	Cardiac interbeat interval (time between successive ECG R-waves)
$\bar{V}_{CYC}$	Vessel spatial peak velocity, averaged over one cardiac cycle
V_{MIN}	Minimum diastolic velocity
T_{MIN}	Time of diastolic minimum velocity
A_{MAX}	Maximum acceleration
$T_{A\,MAX}$	Time of maximum acceleration
$V_{A\,MAX}$	Velocity at point of maximum acceleration
V_{MAX}	Maximum systolic velocity
T_{MAX}	Time of maximum systolic peak-velocity
V_{HM}	$V_{HM} = \frac{V_{MAX} - V_{MIN}}{2} + V_{MIN}$
T_{LHM}	Time during early systole at which $V_{peak} = V_{HM}$
T_{FWHM}	Time period during which $V_{peak} \geqslant V_{HM}$
V_{DN}	Minimum velocity in dicrotic notch
T_{DN}	Time of minimum velocity in dicrotic notch
D_{LCC}	Diameter of left CCA, 2 cm proximal to bifurcation
D_{RCC}	Diameter of right CCA, 2 cm proximal to bifurcation
D_{CC}	Average diameter of left and right common carotid arteries

σ () represents the standard deviation in each of the above parameters.

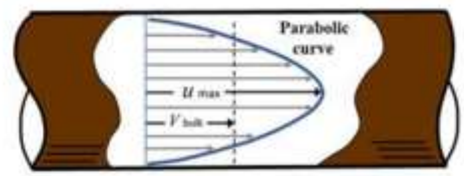
Table 2. Average timing and velocity parameters for 3560 human CCA flow waveforms.

Parameter	Units	Mean	σ_{intra}	σ_{inter}	Max.	Min.
N_{CYC}		52.4		±8.3	110	23
T_{RR}	s	0.917	±0.045	±0.119	1.414	0.601
$\bar{V}_{CYC}$	cm s^{-1}	38.8	±1.5	±7.3	58.7	19.4
V_0	cm s^{-1}	25.9	±3.2	±4.6	43.2	11.1
V_{MIN}	cm s^{-1}	20.9	±2.0	±4.8	36.2	7.0
T_{MIN}	s	0.055	±0.016	±0.014	0.118	0.011
A_{MAX}	cm s^{-2}	2441	±350	±456	8048	1233
$T_{A\,MAX}$	s	0.110	±0.010	±0.010	0.154	0.035
$V_{A\,MAX}$	cm s^{-1}	47.7	±12.3	±7.2	114.2	6.4
V_{MAX}	cm s^{-1}	108.2	±3.8	±17.8	171.2	68.2
T_{MAX}	s	0.153	±0.008	±0.011	0.189	0.118
V_{HM}	cm s^{-1}	64.6	±2.2	±10.2	100.24	39.7
T_{LHM}	s	0.116	±0.005	±0.010	0.151	0.088
T_{FWHM}	s	0.103	±0.006	±0.024	0.257	0.06
V_{DN}	cm s^{-1}	19.4	±2.9	±6.6	45.6	5.6
T_{DN}	s	0.398	±0.007	±0.016	0.448	0.342
D_{LCC}	cm	0.64		±0.06	0.74	0.52
D_{RCC}	cm	0.63		±0.07	0.75	0.53

σ_{intra} and σ_{inter} are the intra-subject and inter-subject standard deviations for each parameter.

- **Plug Flow Profiles:** Spatially uniform velocity distributions across the vessel radius. This simplification completely **ignores wall friction and initial viscous boundary layer development.**

- **Parabolic Profiles:** Spatially fully developed laminar flow patterns following a quadratic velocity reduction from the centerline maximum to zero at the boundary wall due to the physical enforcement of the no-slip condition.
- **Patient-Specific Waveforms:** In vivo velocity distributions captured directly via Doppler ultrasound or magnetic resonance imaging (MRI), providing the highest diagnostic and predictive fidelity despite clinical noise.



u_{max} = maximum local velocity (center of pipe)
V_{bulk} = average bulk velocity

6. Mesh Independence and Verification: The GCI Method

To verify that the numerical results are independent of mesh density, a Grid Convergence Index (GCI) analysis was performed using three progressively refined meshes: coarse, medium, and fine. The GCI method, based on Richardson Extrapolation, provides a quantitative estimate of the discretization error associated with the computational mesh and determines whether the numerical solution has reached the asymptotic range of convergence. This procedure is widely accepted as a standard verification technique in Computational Fluid Dynamics (CFD).

The refinement ratios between successive meshes were calculated using:

$$r21 = \frac{h21}{h1} \qquad r32 = \frac{h32}{h2}$$

where:

- h1 = Fine mesh grid size
- h2= Medium mesh grid size
- h3= Coarse mesh grid size

Mesh Level	**Grid Size, h (mm)**	**Peak Velocity, ϕ (m/s)**
Fine	0.25	0.707
Medium	0.33	0.705
Coarse	0.45	0.700

The peak velocity magnitude at the stenotic region was selected as the monitoring parameter because it represents the location of the highest velocity gradients and is therefore the most sensitive parameter to mesh refinement. Convergence of this parameter ensures accurate prediction of flow acceleration and pressure losses within the stenosis.

Apparent Order of Convergence

The apparent order of convergence was determined using:

$$p = \frac{1}{r21} \left| ln \left| \frac{\in_{32}}{\in_{21}} \right| + q(p) \right|$$

where q(p) is a correction factor accounting for unequal refinement ratios:

$$q(p) = \ln \left(\frac{r^{p}{}_{21} - 1}{r^{p}{}_{32} - 1} \right)$$

7. Treatment

7.1 Finite Element Analysis of Carotid Stent Expansion Using Balloon Loading

The success of a stent depends not only on its ability to expand adequately but also on its mechanical integrity during and after deployment. Excessive deformation may compromise structural stability, while high stress concentrations may lead to material failure or fatigue over time.
The purpose of this finite element analysis (FEA) is to evaluate the mechanical behavior of a carotid stent during balloon expansion by examining:

- Total deformation
- Directional deformation
- Equivalent (von Mises) stress

These parameters provide insight into the stent's flexibility, expandability, and structural reliability.

7.11 Geometry and Model Description

A three-dimensional carotid stent was modeled and positioned within a curved arterial segment to represent realistic anatomical conditions. Unlike simplified straight-vessel models, the curved geometry introduces asymmetrical loading and deformation patterns that better mimic clinical deployment conditions.

Balloon expansion was simulated indirectly through prescribed loading conditions rather than explicitly modeling the balloon structure itself. This approach significantly reduces computational cost while still capturing the essential expansion mechanics of the stent.

7.12 Boundary Conditions and Loading

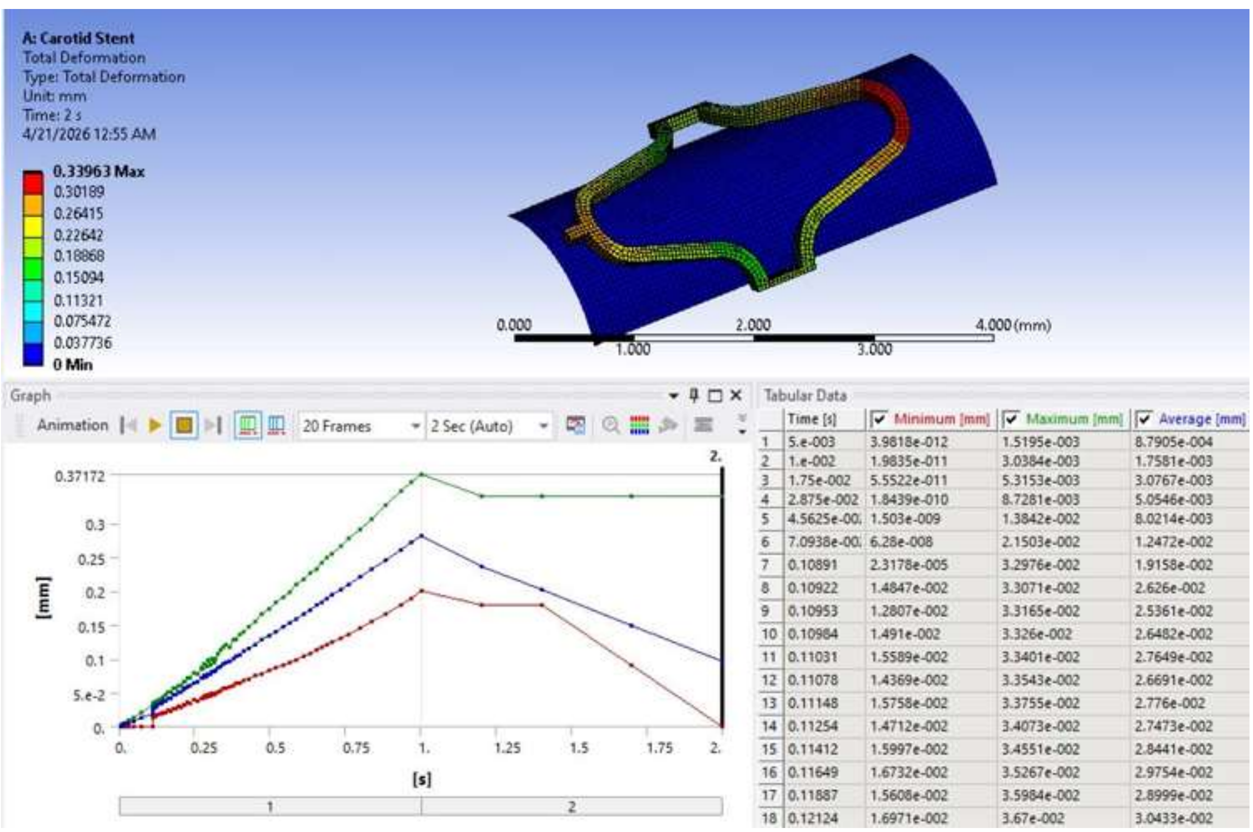
The balloon expansion process was simulated over a period of two seconds using a transient structural analysis.

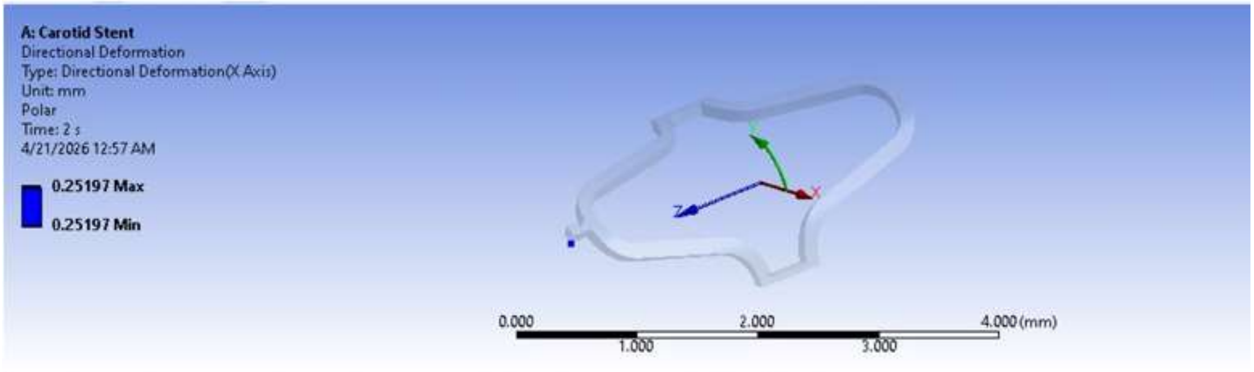
Key loading assumptions included:

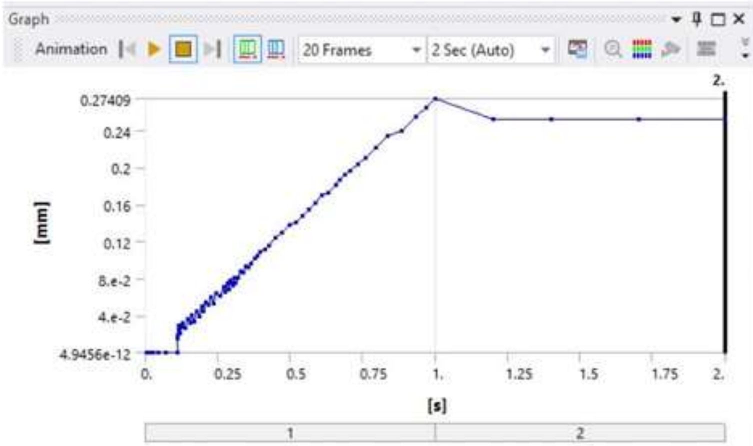
- Gradually increasing expansion load
- Time-dependent displacement
- Structural constraints to prevent rigid-body motion

Applying the load gradually helps replicate the actual balloon inflation process and prevents numerical instability during simulation.

Parameter	Recommended / Optimum Criterion	Obtained Value
Total Deformation	Controlled expansion without instability; typically 0.20–0.50 mm depending on stent geometry and vessel size	**0.3396 mm**
Directional Deformation	Moderate anisotropic expansion expected in curved vessels	**0.252 mm**
Maximum von Mises Stress	Below material yield strength and preferably below fatigue-critical levels	**144 MPa**
von Mises Stress Relative to 316L Stainless Steel Yield Strength (170–290 MPa)	Less than 100% of yield strength	**49.7–84.7% of yield strength**
von Mises Stress Relative to Nitinol Yield Strength (500–700 MPa)	Less than 100% of yield strength	**20.6–28.8% of yield strength**

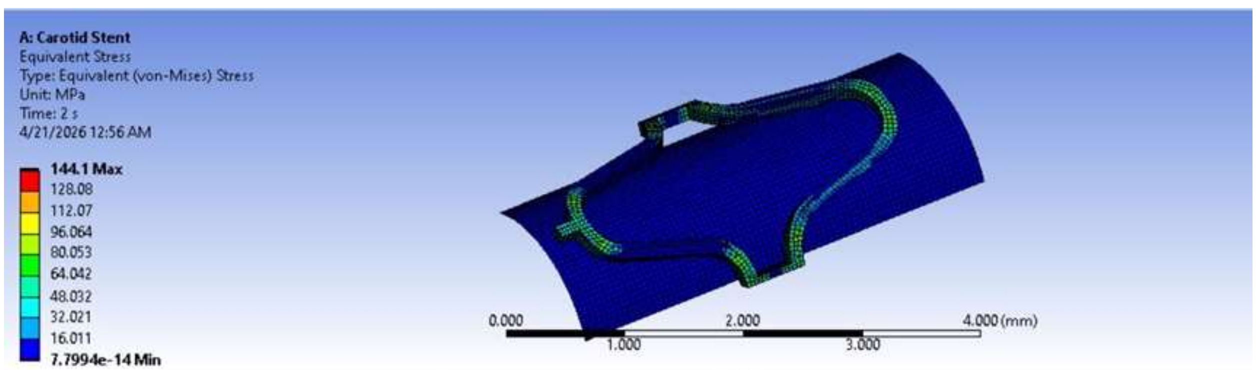


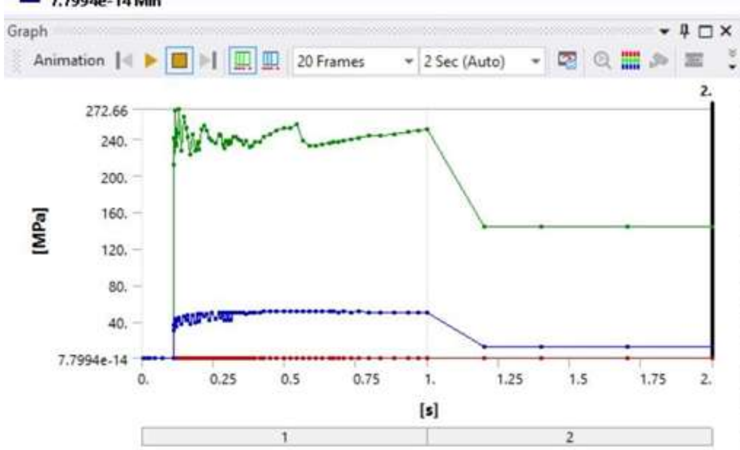




Tabular Data

	Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1	5.e-003	4.9456e-012	4.9456e-012	4.9456e-012
2	1.e-002	2.2251e-011	2.2251e-011	2.2251e-011
3	1.75e-002	6.0869e-011	6.0869e-011	6.0869e-011
4	2.875e-002	2.0025e-010	2.0025e-010	2.0025e-010
5	4.5625e-00:	1.6262e-009	1.6262e-009	1.6262e-009
6	7.0938e-00:	6.7936e-008	6.7936e-008	6.7936e-008
7	0.10891	2.5096e-005	2.5096e-005	2.5096e-005
8	0.10922	1.8413e-002	1.8413e-002	1.8413e-002
9	0.10953	1.5236e-002	1.5236e-002	1.5236e-002
10	0.10984	1.8672e-002	1.8672e-002	1.8672e-002
11	0.11031	2.5178e-002	2.5178e-002	2.5178e-002
12	0.11078	1.817e-002	1.817e-002	1.817e-002
13	0.11148	2.2322e-002	2.2322e-002	2.2322e-002
14	0.11254	2.3002e-002	2.3002e-002	2.3002e-002
15	0.11412	2.5146e-002	2.5146e-002	2.5146e-002
16	0.11649	2.9613e-002	2.9613e-002	2.9613e-002
17	0.11887	2.1274e-002	2.1274e-002	2.1274e-002
18	0.12124	2.5833e-002	2.5833e-002	2.5833e-002





Tabular Data

	Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1	5.e-003	1.2311e-010	1.1994e-007	2.3026e-008
2	1.e-002	1.5731e-009	5.0558e-007	7.709e-008
3	1.75e-002	3.8109e-009	1.399e-006	1.8994e-007
4	2.875e-002	9.1568e-009	4.6272e-006	5.6018e-007
5	4.5625e-00:	3.4549e-008	3.7702e-005	4.1197e-006
6	7.0938e-00:	5.3899e-008	1.5789e-003	1.6792e-004
7	0.10891	8.3171e-008	0.58489	6.1973e-002
8	0.10922	8.3413e-008	240.62	36.306
9	0.10953	8.3655e-008	211.5	30.932
10	0.10984	8.3898e-008	242.01	36.725
11	0.11031	8.4261e-008	238.31	39.746
12	0.11078	8.4624e-008	235.13	33.341
13	0.11148	8.517e-008	234.68	38.399
14	0.11254	8.5987e-008	212.67	35.76
15	0.11412	8.7215e-008	233.04	39.699
16	0.11649	8.9057e-008	271.09	42.242
17	0.11887	9.09e-008	235.16	34.66
18	0.12124	9.2744e-008	232.18	39.881

7.2 Geometrical Configuration of the Stent within the Stenosed Arter

To evaluate the effectiveness of stent deployment in restoring blood flow through a stenosed artery, a finite element structural analysis was conducted using ANSYS Workbench. The objective of this stage was to investigate the mechanical behavior of the stent during expansion and its interaction with the surrounding arterial wall and atherosclerotic plaque. A realistic geometrical model was developed, and appropriate loading conditions were applied to simulate the deployment process. The analysis provides insight into stress distribution, deformation characteristics, and the ability of the stent to reopen the narrowed arterial lumen while maintaining structural integrity.

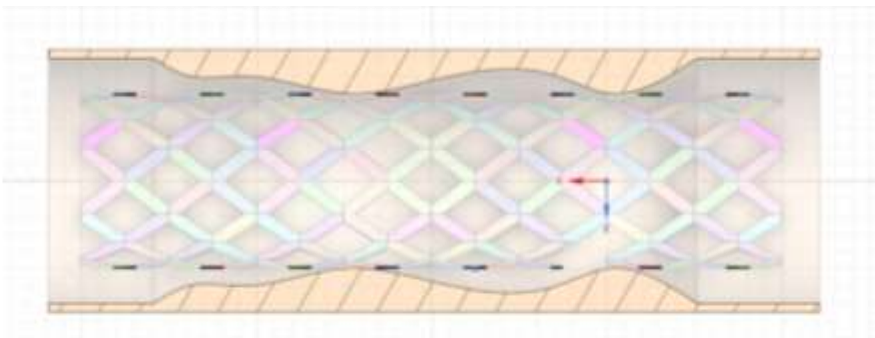


Figure 7.1 illustrates the geometrical model used for the stent deployment simulation. The vascular stent, designed in SolidWorks and imported into ANSYS, is positioned within the stenosed arterial segment. The surrounding artery includes an atherosclerotic plaque responsible for the localized lumen narrowing. This assembled model was used as the initial configuration for the finite element analysis, where pressure and displacement boundary conditions were applied to the inner surface of the stent to simulate radial expansion, plaque compression, and restoration of the arterial lumen.

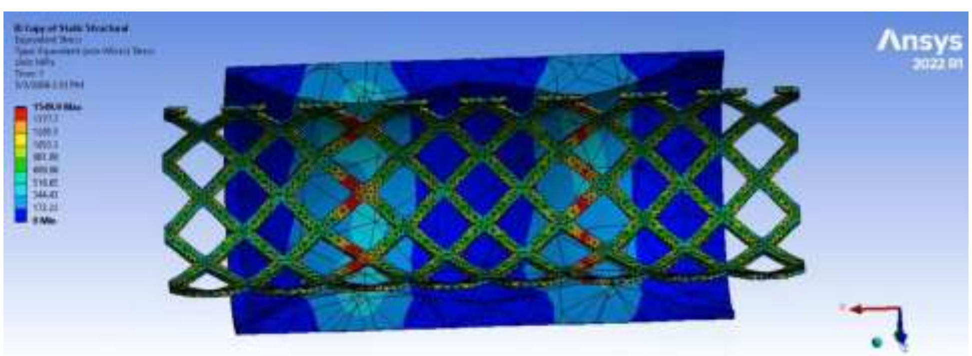


Figure 7.2 demonstrate the expanded stent configuration together with the corresponding stress contours. Areas of higher stress concentration are primarily observed near the connecting crowns and curved regions of the stent struts, where deformation is greatest during expansion. The successful expansion of the stent indicates its ability to restore the lumen diameter and reduce the severity of the stenosis while maintaining structural integrity.